

UTKARSH SARAOGI

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EDUCATION

Northeastern University, Boston | MS in Computer Science

Sep. 2025 - Present

Coursework: MLOps, NLP, Data Mining, Database Management Systems, Algorithms, System Design

PROFESSIONAL WORK EXPERIENCE

Cloud Developer

Aug. 2022 – Aug. 2025

Hewlett Packard Enterprise

Bangalore, India

- Designed and maintained automated data validation framework distributing feature validation scripts across heterogeneous switch fleets simultaneously; covered integration, regression, smoke, and release testing cycles for **P0/P1** critical features; reduced end-to-end validation cycle time by **55%** through workload partitioning and batch scheduling.
- Traced data loss across **15+** microservices ingesting switch telemetry (VLAN/VXLAN segment data, hardware health signals) via CrowdStrike, following records through enricher, predictor, and sink connector stages to pinpoint the failing service and root cause; routed findings to owning teams, reducing MTTR by **40%**.
- Led the triage unit — Built structured training dataset from **4,000+** labeled failure scenarios across categorized failure types like feature, infrastructure, server and script failures from years of debugging logs; designed annotation schema for an ML-powered triage assistant that improved automated failure classification accuracy by **15%**.
- Developed Python automation scripts for structured data collection, log parsing, and feature extraction across the Switch Data Monitoring platform; standardized failure event schemas across heterogeneous switch models enabling consistent feature engineering for downstream ML training datasets.
- Owned Jenkins CI for the platform's test suite and lab environments; maintained validation coverage as GraphQL APIs and deployed services drifted from test expectations, fixing scripts to match live contracts and catching schema mismatches before release rather than in production.

Software Engineer (Intern)

Jan. 2022 – Jul. 2022

Hewlett Packard Enterprise

Bangalore, India

- Wrote and executed automated feature validation scripts integrated into the CI pipeline for the Switch Data Monitoring platform, covering regression and smoke test cycles across **10+** switch models; eliminated recurring production rejections caused by malformed telemetry payloads.
- Performed analysis on failure log extraction scripts by cross-referencing **500+** historical debugging logs for accuracy; authored testing and onboarding documentation adopted by **3** incoming team members, improving platform observability across the switch monitoring stack.

PROJECTS

HindSight - Bitemporal Financial Data Warehouse | Databricks, Lakeflow, Delta Lake, PySpark, Unity Catalog

- Ingested **69** quarters of SEC filings (**181M** facts) through a Lakeflow pipeline with Auto Loader; quarantine-based data contracts held broken row rate to **0.001%**, tag mapping raised revenue coverage from **37%** to **69%**.
- Built a bitemporal core resolving per-fact filing precedence into **21.8M** knowledge intervals across **14,885** companies, enabling as-of queries that return only what was publicly knowable on any date; used liquid clustering since restatements write late-arriving rows into decade-old periods.
- Validated with **6** bitemporal invariants and a dual-implementation equivalence test against a pytest-verified reference; measured look-ahead bias at **3.3%** of top-decile selections differing across **49** quarters.

MarketPulse - Real-Time Market Anomaly Detection Pipeline | Kafka, Spark, Delta Lake, GCP, dbt, Docker

- Built a Kafka-to-GCS Medallion streaming pipeline (PySpark Structured Streaming, Delta Lake ACID transactions), guaranteeing exactly-once processing and 60s anomaly detection latency with correctness guarantees under node failure.
- Created **3** detection signals (Z-score volume spike, price deviation, wash trading pattern) with per-signal confidence scores and alert thresholds; Gold layer loaded to BigQuery; **3** dbt models with **8** passing data quality tests.

SoundStream - Multi-Source Music Data Pipeline | PySpark, dbt, Airflow, Parquet, SQL

- Built a batch ELT pipeline processing **46M+** records from 3 sources using medallion architecture; **60%** storage compression via partitioned Parquet; Bronze, Silver, and Gold layers orchestrated as a DAG in Airflow.
- Implemented star schema with **4** Gold tables; Silver-layer quality checks removed **5,043** duplicates across a **36.7%** cross-source match rate; enabled genre trend analysis and segmentation across **70+** countries.

TECHNICAL SKILLS

Data Engineering	: Kafka, Airflow, dbt, Databricks, Apache Spark, ETL/ELT, Delta Lake, Medallion Lakehouse
Database Systems	: MySQL, MongoDB, PostgreSQL
Cloud & DevOps	: AWS (S3, Lambda, Redshift), GCP (GCS, BigQuery), Docker, Git, CI/CD, Kubernetes
Data Science & ML	: Scikit-learn, PyTorch, Pandas, Matplotlib, A/B Testing, Causal Inference
Programming & Core CS	: Python, SQL, Java, PySpark, System Design